

Industry Data Engineer Roadmap (12 Months)

A structured roadmap from beginner to industry-ready Data Engineer with real-world projects.

Month 1 – SQL Foundations

Skills to Learn:

- SELECT, WHERE, GROUP BY, ORDER BY, joins, aggregations

Free Resources:

- SQLBolt Interactive Tutorial
- Mode SQL Tutorial

Project Idea:

- Project: Analyze a retail sales dataset using SQL queries.

Month 2 – Advanced SQL

Skills to Learn:

- Window functions, CTEs, query optimization, query plans

Free Resources:

- LeetCode SQL Practice
- BigQuery SQL Reference

Project Idea:

- Project: Build analytical queries for an ecommerce analytics dataset.

Month 3 – Python Basics

Skills to Learn:

- Python syntax, functions, loops, file handling, scripts

Free Resources:

- freeCodeCamp Python Course
- Python Official Tutorial

Project Idea:

- Project: Write Python scripts that transform CSV data.

Month 4 – Python for Data Processing

Skills to Learn:

- Pandas, working with JSON/CSV, API ingestion

Free Resources:

- Kaggle Pandas Course
- Pandas Documentation

Project Idea:

- Project: Build a script to pull data from an API and store in a database.

Month 5 – Data Modeling

Skills to Learn:

- Star schema, snowflake schema, dimension & fact tables

Free Resources:

- Kimball Data Warehouse Concepts
- Snowflake Data Warehouse Guide

Project Idea:

- Project: Design a warehouse schema for product analytics.

Month 6 – ETL Pipelines

Skills to Learn:

- ETL vs ELT, ingestion pipelines, transformation pipelines

Free Resources:

- Data Engineering Zoomcamp
- ETL vs ELT Article

Project Idea:

- Project: Build an ETL pipeline that processes CSV data into a SQL warehouse.

Month 7 – Data Warehousing

Skills to Learn:

- OLTP vs OLAP, partitioning, analytics tables

Free Resources:

- BigQuery Documentation
- Snowflake Data Warehouse Guide

Project Idea:

- Project: Create analytical tables for a marketing dataset.

Month 8 – Big Data Concepts

Skills to Learn:

- Distributed computing, Hadoop ecosystem basics

Free Resources:

- Apache Hadoop Overview
- Spark Introduction

Project Idea:

- Project: Process large dataset using distributed processing concepts.

Month 9 – PySpark

Skills to Learn:

- Spark DataFrames, transformations, partitioning

Free Resources:

- Spark SQL Guide
- Databricks Spark Tutorials

Project Idea:

- Project: Build Spark pipeline for processing large log data.

Month 10 – Workflow Orchestration

Skills to Learn:

- Apache Airflow, DAGs, scheduling ETL jobs

Free Resources:

- Apache Airflow Docs
- Astronomer Airflow Tutorial

Project Idea:

- Project: Schedule ETL jobs using Airflow DAGs.

Month 11 – Data Quality & Monitoring

Skills to Learn:

- Data validation, schema checks, pipeline monitoring

Free Resources:

- Great Expectations Docs
- Data Quality Guide

Project Idea:

- Project: Add validation checks to a pipeline using Great Expectations.

Month 12 – Industry Capstone Project

Skills to Learn:

- End-to-end data pipeline

Free Resources:

- Data Engineering Zoomcamp Projects
- freeCodeCamp Data Engineering Project

Project Idea:

- Project: Build full pipeline — ingestion → ETL → warehouse → analytics tables and publish on GitHub.